Unit: 6 Unit 8: Quadratics
Topic: Systems of Quadratic Equations

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. Solve the system: $x^2 + y^2 = 4$, $x + y = 2$
(A) $x = 0, y = 2$
(B) $x = 2, y = 0$
(C) $x = 1, y = 1$
(D) $x = -1, y = -1$
(E) $x = -2, y = 0$

Q2. Solve the system: $x^2 - y^2 = 3$, $x + y = 1$
(A) $x = -1, y = 2$
(B) $x = 1, y = -2$
(C) $x = 2, y = -1$
(D) $x = -2, y = 1$
(E) $x = 0, y = 3$

Q3. What is the graphical representation of the system: $x^2 + y^2 = 1$, $y = x$?
(A) Two intersecting lines
(B) Two parallel lines
(C) A circle and a line
(D) Two circles
(E) A parabola and a line

Q4. Solve the system: $y = x^2 - 2$, $y = -x^2 + 2$
(A) $x = 0, y = 2$
(B) $x = 1, y = -1$
(C) $x = -1, y = -1$
(D) $x = 0, y = 0$
(E) $x = 2, y = -2$

Q5. What is the number of solutions to the system: $x^2+y^2 = 4$, $x^2 + y^2 = 1$?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q6. Solve the system: $y = x^2 + 1$, $y = -x^2 - 1$
(A) $x = 0, y = 1$
(B) $x = 1, y = 2$
(C) $x = 0, y = 0$
(D) $x = -1, y = -2$
(E) No solution

Q7. What is the graphical representation of the system: $y = x^2$, $y = -x^2$?
(A) Two intersecting parabolas
(B) Two parallel parabolas
(C) A parabola and a line
(D) Two circles
(E) A single point

Q8. Solve the system: $x^2 + y^2 = 4$, $x - y = 0$
(A) $x = 0, y = 2$
(B) $x = 2, y = 0$
(C) $x = sqrt2, y = sqrt2$
(D) $x = -sqrt2, y = -sqrt2$
(E) $x = 1, y = 1$

Open Ended Questions (FRQ)

Q9. Solve the system: $y = x^2 - 4$, $y = x + 2$. Show all work and explain your reasoning.

Q10. Graph the system: $x^2 + y^2 = 9$, $y = x^2 - 1$. Label all axes and curves. Describe the number and type of solutions.

Chef's Tips

- Ensure students have access to graph paper for question 10.
- Remind students to show all work for free-response questions.
- Encourage students to check their work by plugging solutions back into the system.